

Problem

If in Example 1.2, $q = (10 - 10e^{-2t})$ mC, find the current at $t = 1.0$ s.

Example 1.2:

The total charge entering a terminal is given by $q = 5t \sin 4\pi t$ mC. Calculate the current at $t = 0.5$ s.

Solution:

$$i = \frac{dq}{dt} = \frac{d}{dt} (5t \sin 4\pi t) \text{ mC/s} = (5 \sin 4\pi t + 20\pi t \cos 4\pi t) \text{ mA}$$

At $t = 0.5$,

$$i = 5 \sin 2\pi + 10\pi \cos 2\pi = 0 + 10\pi = 31.42 \text{ mA}$$

Step-by-step solution

Step 1 of 1

Consider the following data:

The total charge entering a terminal, $q = (10 - 10e^{-2t})$ mC

The time $t = 1.0$ s

An alternating current is a current that varies sinusoidal with time.

$$\begin{aligned} i &= \frac{dq}{dt} \\ &= \frac{d}{dt} (10 - 10e^{-2t}) \\ &= \frac{d}{dt} (10) + \frac{d}{dt} (-10e^{-2t}) \\ &= 0 + (-10)(-2)e^{-2t} \\ &= 20e^{-2t} \end{aligned}$$

At $t = 1.0$ s, calculate the current.

$$\begin{aligned} i &= 20e^{-2t} \\ &= 20e^{-2(1)} \\ &= 2.707 \text{ mA} \end{aligned}$$

Therefore, the current at $t = 1.0$ s is 2.707 mA.